1.7 ONE-TAILED AND TWO-TAILED TESTS

In some hypotheses testing problem, elements of the critical region may be identified by a rejection rule of the type $t(\underset{\sim}{X}) \geq c$. In this case, $P(t(\underset{\sim}{X}) \geq c)$ will be the area, which falls at the right end (Figure1.1) under the curve representing the sampling distribution of $t(\underset{\sim}{X})$. The statistical test defined by this kind of critical region is called **right-tailed test**.

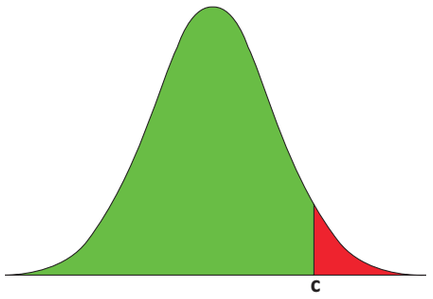


Figure 1.1. Right-tailed Test

On the other hand, suppose that the rejection rule $t(\underset{\sim}{X}) \leq c$ determines the elements of the critical region. Then, $P(t(\underset{\sim}{X}) \leq c)$ will be the area, which falls at the left end (Figure.1.2) under the curve representing the sampling distribution of $t(\underset{\sim}{X})$. The statistical test defined by this kind of critical region is called **left -tailed test**.

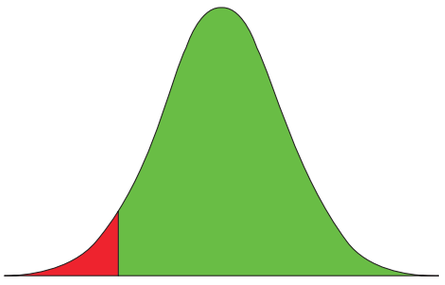


Figure 1.2 Left-tailed Test

The above two tests are commonly known as **one-tailed tests**.

Note 9: It should be noted that the sampling distribution of $t(\underset{\sim}{X})$ need not be with symmetric shape always. Sometimes, it may be positively or negatively skewed.

Example 1.5

Suppose a pizza restaurant claims its average pizza delivery time is 30 minutes. But you believe that the restaurant takes more than 30 minutes. Now, the null and the alternate hypotheses can be formulated as

H_0 : μ = 30 minutes and H_1: μ > 30 minutes

Suppose that the decision is taken based on the delivery times of 4 randomly chosen pizza deliveries of the restaurant. Let X_1, X_2, X_3, and X_4 represent the delivery times of the such four occasions. Also, let H_0 be rejected, when the sample mean exceeds 31. Then, the critical region is

$$\text{Critical Region} = \left\{(x_1, x_2, x_3, x_4) \mid \bar{x} = \frac{x_1 + x_2 + x_3 + x_4}{4} > 31\right\}$$

In this case, $P(\bar{X} > 31)$ will be the area, which fall at the right end under the curve representing the sampling distribution of $\bar{X}$. Hence, this test can be categorized as a right-tailed test.

Suppose that H_0 is rejected, when either $t(\underset{\sim}{X}) \leq a$ or $t(\underset{\sim}{X}) \geq b$ holds. In this case, $P(t(\underset{\sim}{X}) \leq a)$ and $P(t(\underset{\sim}{X}) \geq b)$ will be the areas, which fall respectively at left and right ends under the curve representing the sampling distribution of $t(\underset{\sim}{X})$ (Figure 1.3). The statistical test defined with this kind of rejection rule is known as **two-tailed test**.

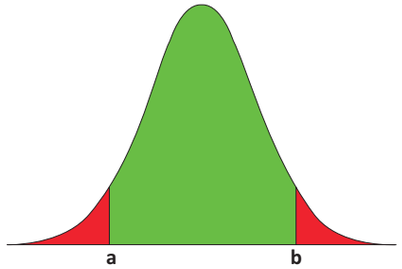


Figure 1.3 Two-tailed Test

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